

18 Data

7 primary questions | Oldest to newest

2026 objectives: 3.6

13 related questions are filed elsewhere in this volume. Find them by selecting this chapter and enabling related topics in [index.html](#).

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CA1-P1 September 2011 Q6	18 marks
CA1-P1 September 2015 Q7	22 marks
CA1-P1 September 2017 Q3	11 marks
CP1-P1 September 2019 Q1	6 marks
CP1-P1 September 2022 Q9	9 marks
CP1-P1 September 2022 Q10	13 marks
CP1-P1 April 2023 Q2	6 marks

CA1-P1 | September 2011 | Q6

Current topic | 18 marks total | Original PDF p4

Related: Ch 17, Ch 29.

- 6** In three years' time, a small provincial town will be celebrating the 750th anniversary of its founding. The organising committee is planning large and expensive celebrations to be held over a weekend during the anniversary year. The committee is worried about the festivities being disrupted or ruined by bad weather.

The part of the country where the town is situated has notoriously unpredictable weather that can cover the full range of possible conditions from mild to extreme.

Data exists covering local, regional and national weather records over the last few centuries. However, the more local the data is, the less complete and accurate it is.

A senior lecturer at the town's college is proposing to use this data in a model that will predict the best date for this event.

- (i) Suggest the criteria that could be used to assess whether the weather on a particular weekend is likely to disrupt or ruin the event. [3]
- (ii) Discuss the extent to which the available data may be appropriate for the prediction model. [9]
- (iii) Outline other considerations that could influence the committee's choice of the date for the celebratory weekend. [2]
- (iv) Set out the difficulties that the committee could face if it tried to fully transfer its financial risks on this project to a third party. [4]

[Total 18]

CA1-P1 | September 2015 | Q7

Current topic | 22 marks total | Original PDF p3

Examiner report and original paper: use the links for this question in index.html.

- 7** (i) List the main uses of data by a financial institution. [4]

A funeral director has provided funeral services in a large city for a number of years. It provides these services through the sale of funeral cost plans, which it only offers directly to individuals.

Under the funeral cost plans individuals pay regular premiums to the funeral director over a period of ten years, or until death if sooner. From the time the funeral cost plan is taken out it guarantees that whenever the individual dies the funeral services will be paid for.

If an individual stops paying premiums while still alive before the ten years is up then the funeral cost plan is cancelled. No payment is made to the individual if the funeral cost plan is cancelled.

- (ii) Describe the data that will be required to set premiums for these funeral cost plans. [10]
- (iii) Compare the data requirements for an insurance company offering whole life assurance policies nationally with those for these funeral cost plans. [8]
- [Total 22]

CA1-P1 | September 2017 | Q3

[Current topic](#) | [11 marks total](#) | [Original PDF p2](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 3**
- (i) List six areas of actuarial work where data would be required. [3]
 - (ii) Describe how actuaries can mitigate or make allowances for the problems that may arise from poor quality data. [8]
- [Total 11]

CP1-P1 | September 2019 | Q1

Current topic | 6 marks total | Original PDF p2

Related: Ch 19.

- 1** A life insurance company has decided to launch a new product in a developing country (Country A) in which it has not operated before.

An actuary working for the life insurance company is carrying out an assessment of life expectancy in Country A for the purposes of projecting future liabilities.

The actuary is considering using past data from the country's statistical authority as a basis for their initial work.

Outline the factors the actuary should take into account when using this data. [6]

CP1-P1 | September 2022 | Q9

Current topic | 9 marks total | Original PDF p4

Examiner report and original paper: use the links for this question in [index.html](#).

- 9** An established life assurance company has been selling term assurance for some years and is about to start selling a new term assurance product.

The new product will be sold to people whose lifestyle and fitness will be monitored by them wearing a wristband.

The wristband collects data that will be shared with the insurance company. Policyholders will then be offered various benefits and rewards from the insurance company depending on the data collected from them.

- (i) Discuss the additional data governance issues that may arise from this new product compared to the existing term assurance business. [6]
- (ii) Outline the main limitations of the wristband data as a means of determining the level of benefits and reward to be given to a policyholder. [3]

[Total 9]

CP1-P1 | September 2022 | Q10

Current topic | 13 marks total | Original PDF p4

Related: Ch 23.

- 10** An insurance company is investigating replacing its existing method for pricing its car rental insurance policies with a more sophisticated approach. It is planning to use publicly and commercially available data as a starting point for pricing these policies.
- (i) Suggest the types of publicly and commercially available data that the insurance company could use. [2]
 - (ii) Discuss why the new data may be an improvement on the existing method used to price for risk. [3]
 - (iii) Outline the problems that may be involved in using this new data. [3]
 - (iv) Discuss how suitable this new data would be for pricing its other motor insurance policies. [5]
- [Total 13]

CP1-P1 | April 2023 | Q2

Current topic | 6 marks total | Original PDF p2

Examiner report and original paper: use the links for this question in [index.html](#).

- 2** An insurance company has a sales promotion on its website. In return for providing personal information, including details of any current or historic illnesses, the applicants will receive a term assurance quotation and be entered into a competition to win a car.
- (i) Describe possible issues an insurance company should consider when asking applicants about their personal data. [3]
 - (ii) Outline what would be included in the company's data governance policy that could resolve the issues in part (i). [3]
- [Total 6]